CARTEYE PITCH DECK

POINT YOUR PHONE.

DETECT PRODUCTS.

GET YOUR BILL

KUNAL | VEDANT KAPIL

THE PROBLEM

Checkout is still a sequential, one-by-one process. Every product needs its own scan — and customers pay the price in wasted time, especially during peak hours.

Today's checkout workflow:

Customers queue while each item is scanned one at a time
Repetitive barcode scanning slows down every transaction
Counter staff perform the same motion hundreds of times daily
Peak-hour friction leads to long lines and poor experience

> The bottleneck isn't the cashier — it's the process. Sequential scanning is a structural inefficiency built into every retail counter.

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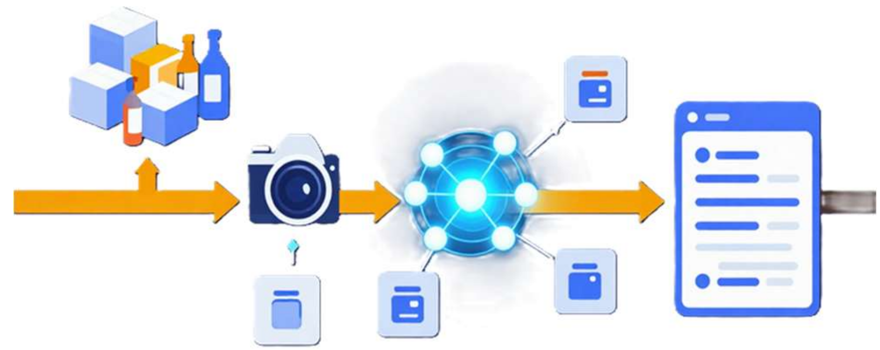
OUR SOLUTION



From Sequential Scanning to Simultaneous Recognition — one camera frame, multiple products, instant bill.

Multiple Products Together

Place all products at once — no individual scanning. CartEye reads the entire basket in a single frame.

One Camera Frame

A single overhead or counter-facing camera captures all items simultaneously — no conveyor belt needed.

Custom YOLO Recognition & Auto Bill

Custom-trained YOLO model detects each product, looks up price, filters duplicates, and generates a complete invoice instantly.

Workflow: Smartphone Camera → AI Detection → Product Recognition → Automatic Cart → Invoice Generated

HOW CARTEYE WORKS

A Layered AI Pipeline from Camera to Invoice

Custom Dataset + Roboflow Annotation

YOLO Training → Best Model Export
Live Camera → Multi-Object Detection
Product DB → Price Lookup + Duplicate Filter
Cart Engine → Invoice → Payment

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WHY A CUSTOM DATASET?

Our custom dataset contains diverse product images captured under varied angles, lighting, and arrangements. The images are annotated with bounding boxes in Roboflow and prepared for YOLOv11 training and validation.

DATASET COMPOSITION:-

70% — Training Set

20% — Validation Set

10% — Testing Set

DATA + AI PIPELINE

REAL-TIME MULTI-OBJECT PRODUCT DETECTION

AI DEMO

PRODUCT CLASSES

Coke , Lays , Bisleri and other items

YOLOv11 inference output: product class + bounding box + confidence score.
Only validated detections proceed to the cart engine.

CONFIDENCE THRESHOLDING

low confidence detections are filtered before cart insertion.

DETECTION OUTPUT

Product Class , Bounding Box
Confidence Score and
Multi-object Localization

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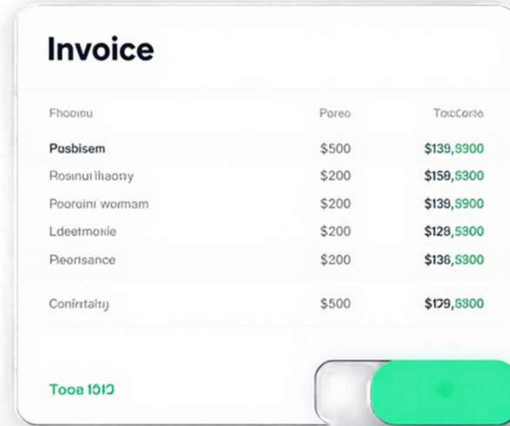
DETECTION → CART → INVOICE



YOLOv11 → Product DB → Price Mapping → Cart Engine → Invoice

CartEye converts validated YOLOv11 detections into structured cart data, automatically mapping products to prices and generating the final invoice.

Detection → Cart Pipeline

YOLOv11 detects the product class → Product DB maps the item to its price → validated items are added to the cart.

Temporal Confirmation

Repeated detections across consecutive frames are validated to prevent duplicate cart entries.

Automated Bill Generation

Confirmed cart items are converted into structured invoice data with product names, quantities, unit prices and totals.

**Invoice data shown is illustrative/demo only. Actual prices and product entries depend on the configured product database at deployment.*

VALIDATION & PERFORMANCE

PRECISION

pending

RECALL

pending

mAP@50

pending

INFERENCE FPS

pending

Custom product images were annotated and augmented in Roboflow for YOLOv11 training and validation.

CHECKOUT BENCHMARK:
Barcode Checkout vs CartEye
Average Checkout Time
Items per Transaction
Time Reduction (%)

Results: Pending Experiment

WHY IT MATTERS

CartEye reduces repetitive scanning through simultaneous multi-product recognition, enabling faster and smarter checkout.

From visual recognition to automated billing, CartEye turns the checkout process into a single intelligent workflow.

Designed impact areas:

1 — FASTER CHECKOUT
Multiple products recognized in a single camera frame.

2 — SMARTER BILLING
AI detection → product mapping → automated cart → invoice.

3 — SCALABLE RETAIL
Smartphone-based vision system designed for future retail automation.

THE FUTURE OF CHECKOUT

FUTURE ROADMAP

— SMARTER VISION

More product classes + improved detection robustness.

— RETAIL INTEGRATION

Inventory synchronization + barcode/RFID support.

— INTELLIGENT SECURITY

Fraud detection + checkout anomaly monitoring.

— EDGE AI

Faster, privacy-focused inference directly on retail devices.

SDGs followed by Carteye:

SDG 9 — Industry, Innovation & Infrastructure

AI-powered computer vision enables smarter and more efficient retail infrastructure.

SDG 12 — Responsible Consumption & Production

Digital billing and data-driven checkout support more efficient retail operations and reduce unnecessary paper-based processes.

KUNAL — TEAM LEADER

VEDANT KAPIL

TEAM THUNDER